RELATIVE TOR DIMENSION OF A COMPLEX ON AN AFFINE MORPHISM EQUALS TOR DIMENSION OVER THE RING

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Allowed standard prerequisites (editorial).

- For a complex of sheaves the tor amplitude in [a, b] can be checked on stalks, comparing the stalks of the sheaf complex with the localized module complex. invoked at proof lines 2577-2587. Converts the sheaf-theoretic tor amplitude condition into the module-theoretic one at every prime.
- A complex of modules has tor amplitude in [a, b] if and only if all its localizations at primes of the base have tor amplitude in [a, b]. invoked at proof lines 2588-2592. Passes from the stalkwise statements at every prime of A back to the global statement for M^bullet over R.
- On a quasi-compact scheme a complex that locally has finite tor dimension has uniform tor amplitude in some [a, b]. used at proof lines 2594-2597. Lets (2) follow from (1) by choosing a uniform amplitude bound.

Source reference. Original lemma 09U9: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 2580-2587 in both directions of (1); the stalk dictionary it is applied through is inside the counted span.

Source reference. Original lemma 0B67: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 2588-2592 to pass from all stalks back to M^bullet over R; the converse assembly is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of affine schemes corresponding to the ring map $R \to A$. Let $M^\bullet$ be a complex of A-modules and let E be the corresponding object of $D(\mathcal{O}_X)$. Then*

(1) *E as an object of $D(f^{-1}\mathcal{O}_S)$ has tor amplitude in $[a, b]$ if and only if $M^\bullet$ has tor amplitude in $[a, b]$ as an object of $D(R)$.*
(2) *E locally has finite tor dimension as an object of $D(f^{-1}\mathcal{O}_S)$ if and only if $M^\bullet$ has finite tor dimension as an object of $D(R)$.*

Proof. Consider a prime $\mathfrak{q} \subset A$ lying over $\mathfrak{p} \subset R$. Let $x \in X$ and $s = f(x) \in S$ be the corresponding points. Then $(f^{-1}\mathcal{O}_S)_x = \mathcal{O}_{S,s} = R_\mathfrak{p}$ and $E_x = M^\bullet_\mathfrak{q}$. Keeping this in mind we can see the equivalence as follows.

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If $M^\bullet$ has tor amplitude in $[a, b]$ as a complex of R-modules, then the same is true for the localization of $M^\bullet$ at any prime of A. Then we conclude by Cohomology, Lemma 09U9 that E has tor amplitude in $[a, b]$ as a complex of sheaves of $f^{-1}\mathcal{O}_S$-modules. Conversely, assume that E has tor amplitude in $[a, b]$ as an object of $D(f^{-1}\mathcal{O}_S)$. We conclude (using the last cited lemma) that $M^\bullet_{\mathfrak{q}}$ has tor amplitude in $[a, b]$ as a complex of $R_{\mathfrak{p}}$-modules for every prime $\mathfrak{q} \subset A$ lying over $\mathfrak{p} \subset R$. By More on Algebra, Lemma 0B67 we find that $M^\bullet$ has tor amplitude in $[a, b]$ as a complex of R-modules. This finishes the proof of (1).

Since X is quasi-compact, if E locally has finite tor dimension as a complex of $f^{-1}\mathcal{O}_S$-modules, then actually E has tor amplitude in $[a, b]$ for some a, b as a complex of $f^{-1}\mathcal{O}_S$-modules. Thus (2) follows from (1). $\square$